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Summary

Computer Engineering graduate with a versatile background spanning **backend development**, **mobile/frontend development**, and applied **machine learning**. Experienced in architecting backend systems with **Node.js**, **Express**, and **MongoDB**, building Android applications with **Kotlin** and **Jetpack Compose**, and developing responsive web interfaces with **Nuxt**, **TypeScript**, and **Tailwind CSS**. Proven ability to **lead cross-functional teams** as both engineer and product manager, with strong skills in modular architecture, REST API integration, and end-to-end product delivery across mobile, web, and AI domains.

Education

Institut Teknologi Sepuluh Nopember, Computer Engineering Sept 2021 – Sept 2025

- GPA: 3.53/4.0
- Thesis: developed a photorealistic virtual musician avatar in **Unreal Engine 5** and **Metahuman**, integrating motion capture (Move AI, Rokoko, DeepMotion) and facial tracking for real-time synchronized performance, evaluated via a custom **MediaPipe**-based landmark analysis pipeline.

Experience

Full Stack Trainee, Korea-ASEAN Digital Academy (KADA) - Indonesia (Bootcamp) Jun 2026 – Aug 2026

- Led a **6-person cross-functional team** as Backend Developer and PM to build Artisan Inventory, a full-stack F&B inventory management system; built **6 core backend modules** (Auth, Inventory, Menu, Production Plan, Selling, Report) using **Node.js**, **Express**, and **MongoDB** with atomic multi-collection transactions, and exposed inventory data to power the team's **AI chatbot** for natural-language stock queries.
- Independently designed and built an **AI-powered running analytics app**, extracting workout data via **OCR** from Huawei Band 11 Pro screenshots and using the **Gemini API** to generate personalized training plans, fitness reports, and pace zones, with one-click sync to **Google Calendar**.
- Completed the full KADA curriculum (Web Development, Backend Development, DevOps & CI/CD, Data Analysis, AI & LLM Applications, UI/UX Design) and ranked **top 5 in class** with a **95** score on the post-assessment.

Junior Frontend Developer, PT. Kreasi Online Indonesia – Indonesia (Internship) December 2025 – Juni 2026

- Developed and maintained production-ready web applications using **Nuxt.js** framework with modular component architecture.
- Integrated multiple **RESTful APIs** across profile, my feedback, change password, announcement, landing pages and switch user, ensuring proper data handling and error validation.
- Collaborated with backend engineers to validate API contracts using Postman, while contributing to debugging with analyst on **SIT** and **UAT**, deployment within an agile development workflow use **Basecamp**.

Mobile Android Developer, PT. Adinata Charming Emmanuel – Surabaya, Indonesia Jul 2024 – Feb 2025
(Freelance)

- Developed **Ace Member**, a membership & promo management app for 4 user roles (Admin, Mitra, Resepsionis, Member).
- Implemented **20+ REST API integrations** (Auth, Promo, Merchant, File Upload, Token) using Kotlin, Clean Architecture (MVVM) with modular XML-based UI following Figma designs.
- Conducted **40+ end-to-end test scenarios** across multiple roles, ensuring feature compliance and reducing bugs before release, while delivering the MVP within the project timeline using Scrum (~90% sprint completion across 10+ sprints).

Mobile Application Developer, Bangkit Academy – Remote (Indonesia) (Study Independent)

Feb 2024 – Aug 2024

- Developed **TerraVision**, a real estate investment Android platform, as a **Mobile Developer** in a 6-person cross-functional team, achieving an **A grade (91.6/100)** in the Capstone Project.
- Built Android apps using **Kotlin, Jetpack Compose, Room, and MVVM**, integrating REST APIs, local databases, and on-device **TensorFlow Lite models**.
- Graduated from **Bangkit Academy 2024** (Google, GoTo, Traveloka) with final program score **89.08/100 (Grade A)** and **100% attendance**.

Projects

BAKI: Sport Motion Detector Based on ML Kit in Mobile App

- Led a 3-member team (Mobile, ML, Backend) in developing **BAKI**, an Android fitness app with AI-powered form correction, built under Scrum with 25 tasks and ~90% completion.
- Engineered the core engine for **real-time workout classification** (push-up, sit-up, squat, lunges) using ML Kit pose landmarks (33 points), a KNN classifier, and EMA smoothing, integrated with an automatic repetition counter.
- Delivered supporting features including OTP-based authentication, Daily Plan scheduling, workout history, an auto rest timer, and two performance modules – **Anticipation Test** (ball-drop timing) and **Reflex Test** (audio-based, 6-level benchmark).
- Implemented 40+ XML layouts with Material Design and **CameraX** live tracking, following **Clean Architecture (MVVM + Dagger Hilt)** for modularity and maintainability.

DogVision: On-Device Dog Breed Classifier Using PyTorch and TFLite

- Built a custom **CNN with residual blocks from scratch (PyTorch)** and applied Mixup augmentation on a 10-breed subset (~20K images) of the Stanford Dogs dataset.
- Achieved **70% validation accuracy from scratch** and up to **85–86% with transfer learning** (MobileNetV2 on mobile, ResNetV2 on web).
- Deployed cross-platform via **PyTorch → ONNX → TensorFlow → TFLite**: an Android app with real-time offline inference and local history (Room DB), plus a Flask + TensorFlow web version.

Skills

Languages: Kotlin, Java, Python, TypeScript, JavaScript, SQL, XML

Frameworks & Libraries: Node.js, Express, Mongoose, Next.js, React, Nuxt, Vue, Jetpack Compose, Android Jetpack (Navigation, Room, ViewModel, LiveData), Dagger Hilt, Tailwind CSS, Zustand, Framer Motion, PyTorch, TensorFlow Lite, OpenCV, Keras

Tools & Platforms: MongoDB, Vercel, Android Studio, Firebase, Git, Postman, CameraX, ML Kit, ONNX, CSV Processing, Gradle, CircleCI, GitHub, Jupyter Notebook, Google Drive API, Supabase, Netlify

Concepts: Clean Architecture (MVVM), REST API Design, Database Transactions (ACID), Agile/Scrum, On-Device Machine Learning, Real-Time Image Processing

Certification

- **Training on KADA** – Komdigi, 2026
- **Becoming an Android Developer Expert** – Dicoding, 2024
- **Applying Machine Learning for Android** – Dicoding, 2024